EFY TESTED
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GETTING TO KNOW IC 555 Through Experiments

Part 1

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There are at least twelve simple experiments that can be performed using IC 555 and simultaneously learn the theory and digital logic behind its operation. Here you will find the first six experiments and the rest will be provided in second part of this article to follow soon.

Although these circuits are experimental and non-conventional, they help in understanding the IC that comprises a voltage divider, trigger comparator, threshold comparator, SR flip-flop, discharge transistor, reset transistor, and the output stage. These also help in understanding its three modes of operation: monostable, astable, and bistable.

All the experiments are simple and can be performed on a breadboard. A power supply of 5V to 9V DC may be used for all the circuits (although, according to its data sheet 4.5V to 15V can be used). The operation of all the twelve experiments is confirmed by the glowing of LEDs.

In the first experiment only a part of the internal circuitry of the IC is used. In second, third, and fourth experiments output pin 3 of the IC 555 is not used but the monostable, astable, and bistable operations using internal two comparators and an SR flip-flop occur, as they normally occur, and their performance is demonstrated by the LEDs. In the fifth and sixth, two 555 ICs are used to create an interesting situation. The experiments make the learning of IC 555 an enjoyable experience.

Experiment 1

Circuit diagram for conducting the first experiment is shown in Fig. 1. It has three sections: 1(a), 1(b), and 1(c).

For experiment 1(a), pin numbers 4 and 8 of the IC are connected to the (5V to 9V) power supply Vcc, while pins 1, 2, and 6 are grounded. The LED is connected between pin 3 and ground through current-limiting resistor R1. When the circuit is powered, LED1 turns on and glows continuously. Here only the lower comparator and output stage of the IC are used.

For experiment 1(b), pin numbers 4, 6, and 8 of the IC are connected to the power supply; pin 2 is connected to the power supply through resistor R2. Pin 1 is grounded. LED2 is connected between Vcc and pin 7 through current-limiting resistor R3. When the circuit is powered, LED2 turns on and glows continuously. Here only the upper comparator and the discharge transistor of the IC are used.

For experiment 1(c), pin numbers 4 and 8 of the IC are connected to the power supply; pin 2 is connected to Vcc through resistor R4. Pins 1 and 6 are grounded. LED3 is

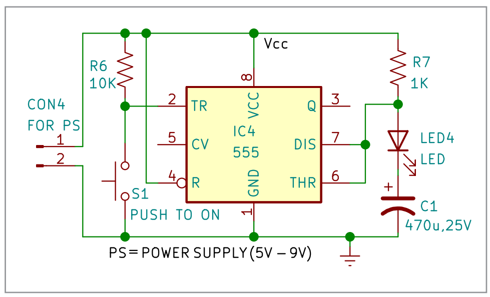

Fig. 2: Circuit diagram for experiment 2

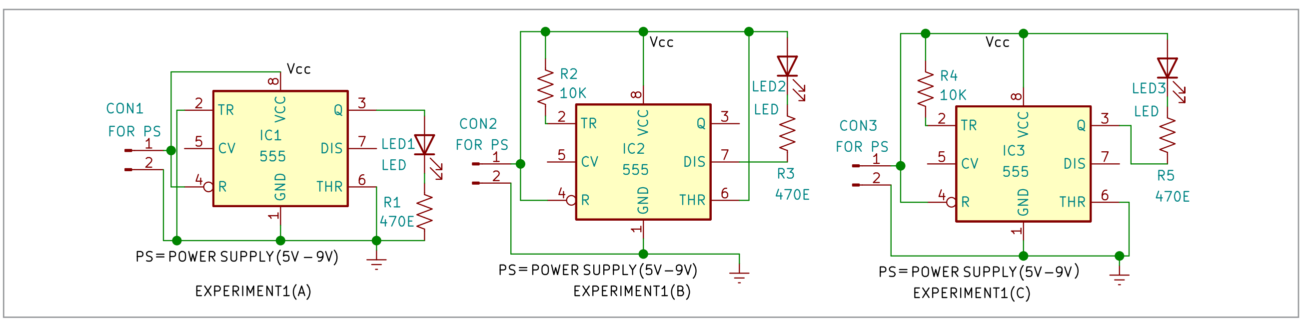

Fig. 1: Circuit diagram for experiment 1